CSBP711 ADVANCED ARTIFICIAL INTELLIGENCE · PAPER CRITIQUE + COURSE PROJECT · UAEU · FALL 2026

Two reality checks on children's pneumonia X-rays

Part 1 · Paper critique: does a Vision Transformer really beat CNNs? (Singh et al., Scientific Reports 14:2487, 2024)
Part 2 · Course project: can an LLM write a classifier's weights itself, with no training at all?

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Same dataset and leakage-free pipeline as our Assignment 1 (Kermany pediatric chest X-rays, patient-grouped splits, official test folder held out).
About 18 minutes: critique first, then the project.

The question: two numbers on the same images, 11 points apart

Assignment 1 gave us a fair, patient-grouped result. The paper reports something much higher. One of two things is true.

OUR ASSIGNMENT 1

86 %

accuracy of the best model (ResNet-50), AUROC 0.971. Patient-grouped split, duplicates removed, the dataset's own test folder, two seeds.

SINGH ET AL. 2024

97.6 %

accuracy of a Vision Transformer (DeiT-Base), AUC 0.966. Random 80/10/10 split, one run, fifteen CNNs reported at 76–94 %.

Either the transformer is genuinely better, or the number comes from how it was evaluated.

Our pipeline can separate the two: run their model under our protocol, and their protocol on our models.

The paper in one line: problem, data, model, claim

Singh, Kumar, Kumar, Verma, Abhishek & Selvarajan. Scientific Reports 14:2487, January 2024. No code released.

1 → 2 → 3 → 4 → 5

Problem

Is this child's frontal chest X-ray NORMAL or PNEUMONIA?

Data

Kermany et al. 2018: 5,856 images, Guangzhou Women and Children's Medical Center, 73 % pneumonia

Split

Random 80 / 10 / 10 by image: 4,684 train, 586 val, 586 test. Patients not mentioned.

Model

DeiT-Base/16 at 224 px, ImageNet weights, fully fine-tuned: Adam, LR 1e-5 × 0.995/epoch, batch 16, 30 epochs

Claim

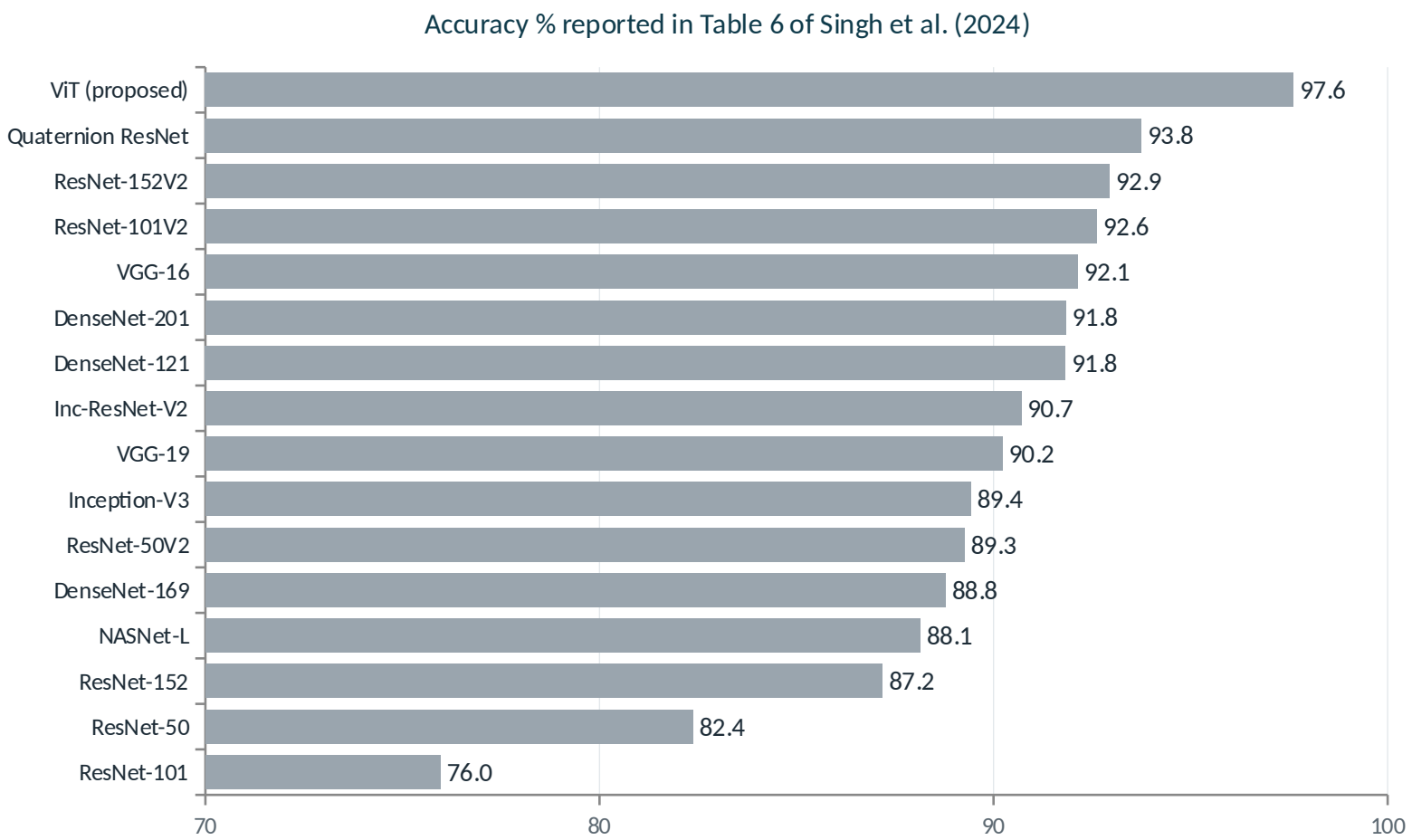
97.61 % accuracy, sens. 0.949, spec. 0.981, AUC 0.966: "a great improvement over all other architectures"

What they offer as evidence

One training run on one split. Accuracy with a Wald 95 % CI, a confusion matrix, ROC and precision–recall curves, and Table 6: fifteen CNN architectures scoring 76–94 % under what the text calls the same hyperparameters.

Their evidence: Table 6, fifteen CNNs against one transformer

Accuracy as reported by Singh et al. The ViT is 4.7 points above the best CNN and 15 points above ResNet-50.



What the table says

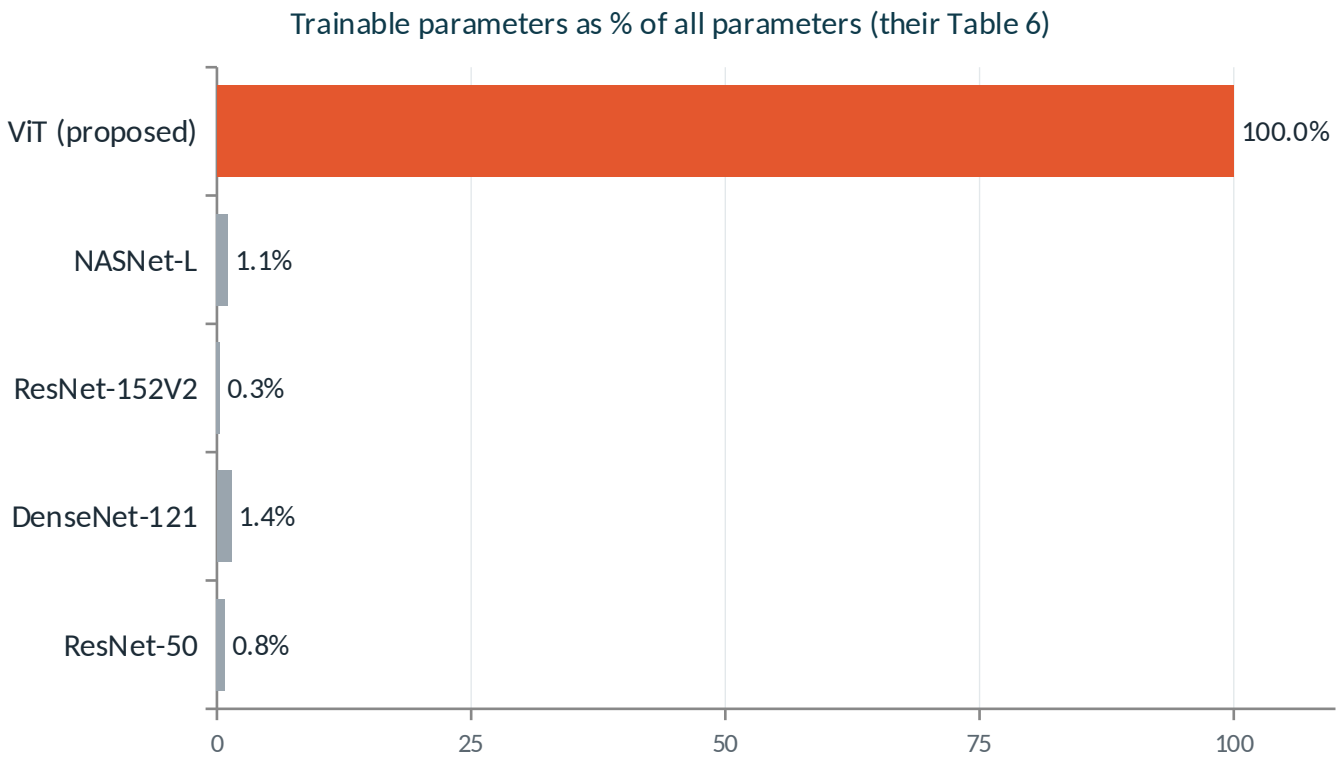
The transformer wins by a wide margin over every CNN, including architectures far larger than it (NASNet-L, 85 M).

What the table also says

A column most readers skip: trainable parameters. ResNet-50: 200,706. DenseNet-121: 100,354. ViT: 85,800,194. Next slide.

Weak point 1: CNN baselines frozen, transformer fully trained

Share of each model's parameters that were trained, from the two parameter columns of their Table 6.



0.8 %

of ResNet-50 was trained: a 200 k head on a frozen 23.6 M backbone

100 %

of the ViT was trained: all 85.8 M parameters

So the comparison is

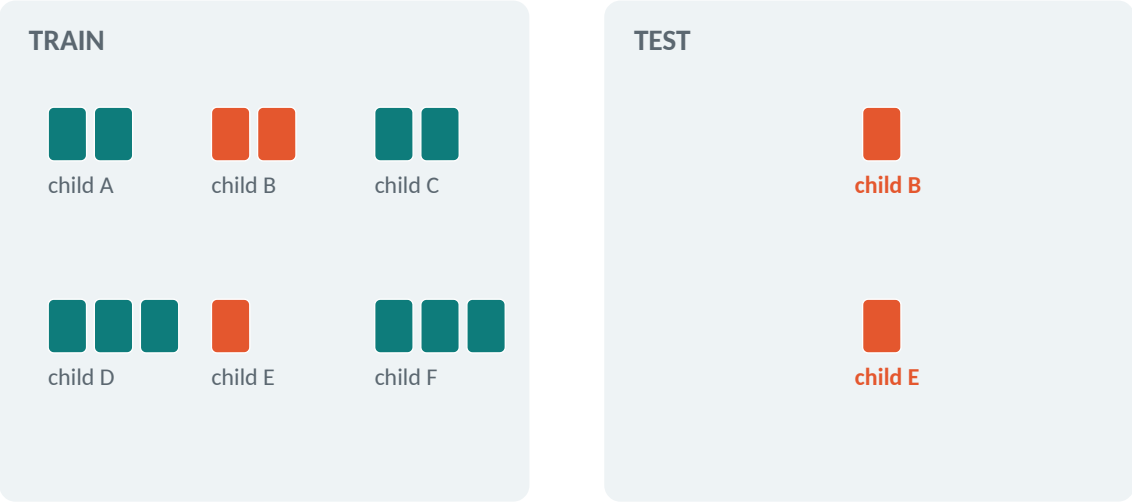
fine-tuning vs feature extraction, not transformer vs CNN. "All hyperparameters were kept the same" is true of the optimiser, not of the training.

Also: a 'Ref.' column sits beside every baseline, so some CNN numbers may be cited rather than run. The paper does not say.

Weak point 2: a random split by image puts one child on both sides

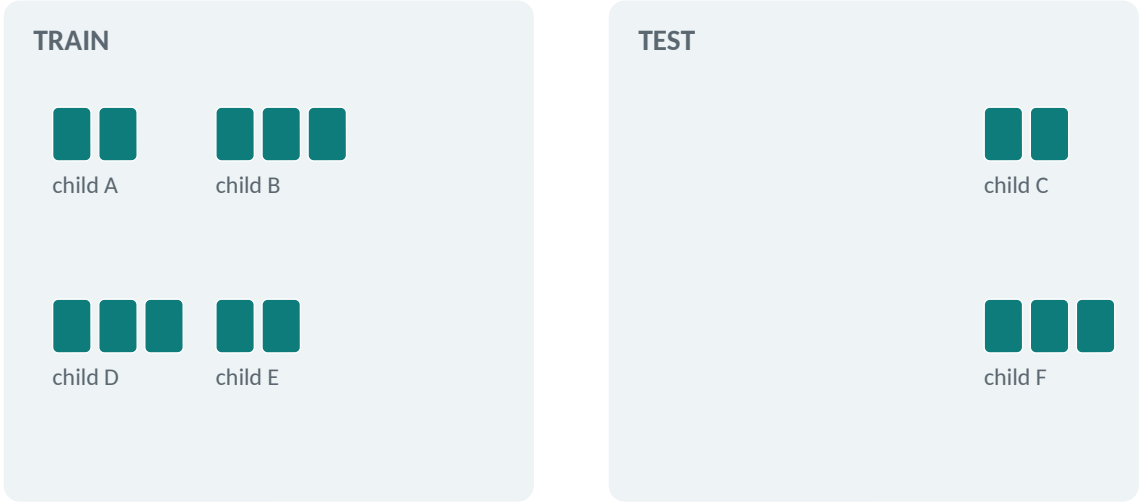
1,155 children in this dataset have more than one X-ray, up to 30. The paper never mentions patients.

Their split: random by image



A scan of child B and child E is in the test set while their other scans are in training. The model can recognise the child instead of the disease.

Our split: whole children



Every scan of a child is on one side only. The model must generalise to children it has never seen.

58 %

of test images have a sibling scan in training when we reproduce their split (336 of 583)

0 %

under our patient-grouped split

And their test set comes from the training pool

The dataset's own test folder is a different collection: in A1 every model scored ≥ 0.99 AUROC on validation but 0.90–0.97 on it. A random 10 % never faces that shift.

WHAT THE EVIDENCE SUPPORTS, BEFORE WE RUN ANYTHING

The claim is larger than the numbers

Supported

"A fully fine-tuned DeiT-Base reaches about 98 % accuracy on a random image-level split of this dataset."

Not supported

"A Vision Transformer outperforms CNN architectures on pediatric pneumonia detection."

Strengths, for balance: a clear and falsifiable claim, a public dataset, a complete hyper-parameter table, a confusion matrix we can recompute from, and a cost note. That is what makes the claim testable.

Reality check design: two models × two protocols, one pipeline

Everything from Assignment 1 is unchanged. We added their model, and a switch that reproduces their split.

	A1 protocol patient-grouped split, official test folder (n = 618)	Paper protocol random 80/10/10 split by image (n = 583)
ViT DeiT-B/16 (their model)	**NEW · fair test of their model** Does the ViT beat ResNet-50 when both are fine-tuned under one protocol?	**NEW · reproduce their number** Do we get their 97.6 % with their model and their split?
ResNet-50 (our A1 winner)	**A1 result** 0.971 AUROC, 86.2 % accuracy. Our reference point.	**NEW · their split on our model** Does a fine-tuned ResNet-50 also reach ~97 % under their split?

Gaps we could not close: their 30 epochs (we keep 12), plain resize (we pad), Adam (we use AdamW). All three can only help their number; the report lists them.

Their model, faithfully

timm deit_base_patch16_224, ImageNet weights, fully fine-tuned, LR 1e-5 as in the paper. 85.8 M params.

Their split, faithfully

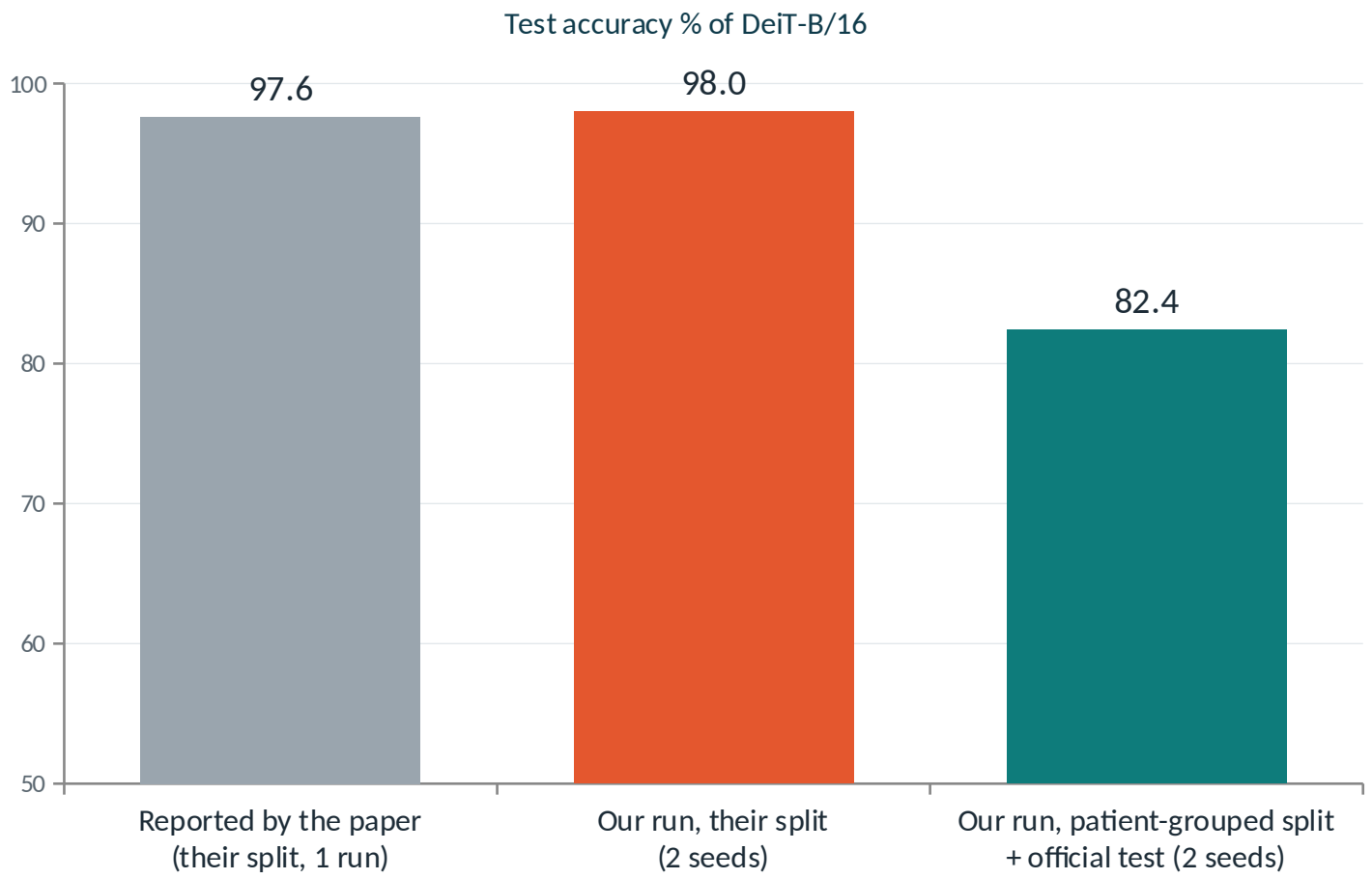
Pool the 5,824 kept images, random label-stratified 80/10/10 by image: 4,659 / 582 / 583. Leakage measured: 58 %.

Held fixed

Grayscale, pad-to-square, 224 px, same augmentation, AdamW, 12 epochs, best-val-AUROC checkpoint, seeds 42 + 43, one T4.

Result 1: their 97.6 % reproduces, and it depends entirely on the split

The same ViT, the same pipeline. Only the split changes.



98.0 %

our ViT under their split; they report 97.6 %. Specificity 0.97 vs their 0.98. Reproduced.

82.4 %

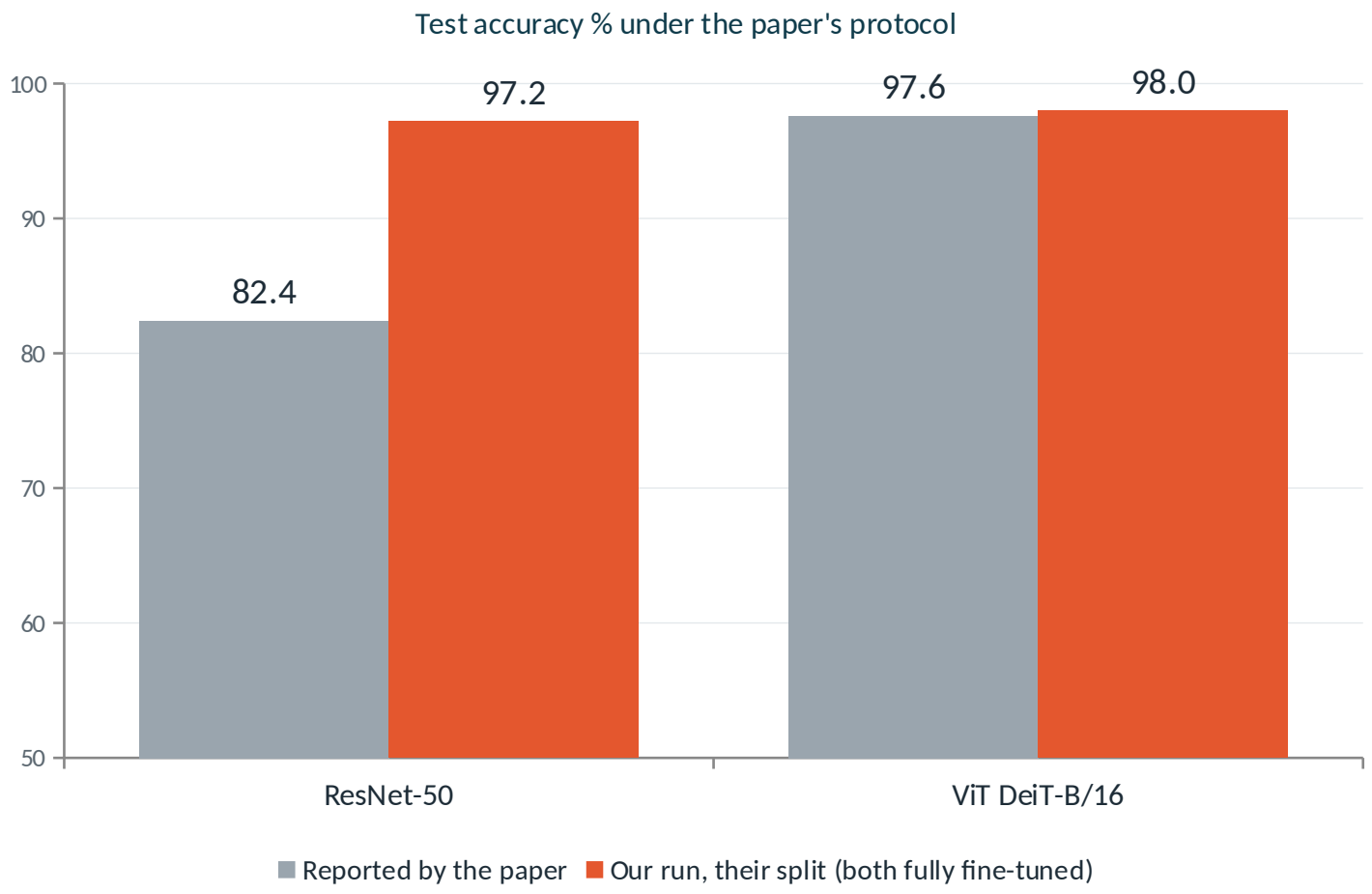
the same ViT under the fair protocol. Specificity 0.53.

Read it as

The number is real, and we got it. But it needs a test set that shares children with training and is drawn from the same collection. Remove both, and 15.6 points disappear. The model did not change; the evaluation did.

Result 2: under their split, a plain ResNet-50 gets the same number

The 15-point gap in their Table 6 is the frozen backbone, not the architecture.



97.2 %

fine-tuned ResNet-50 under their split (two seeds)

+14.8

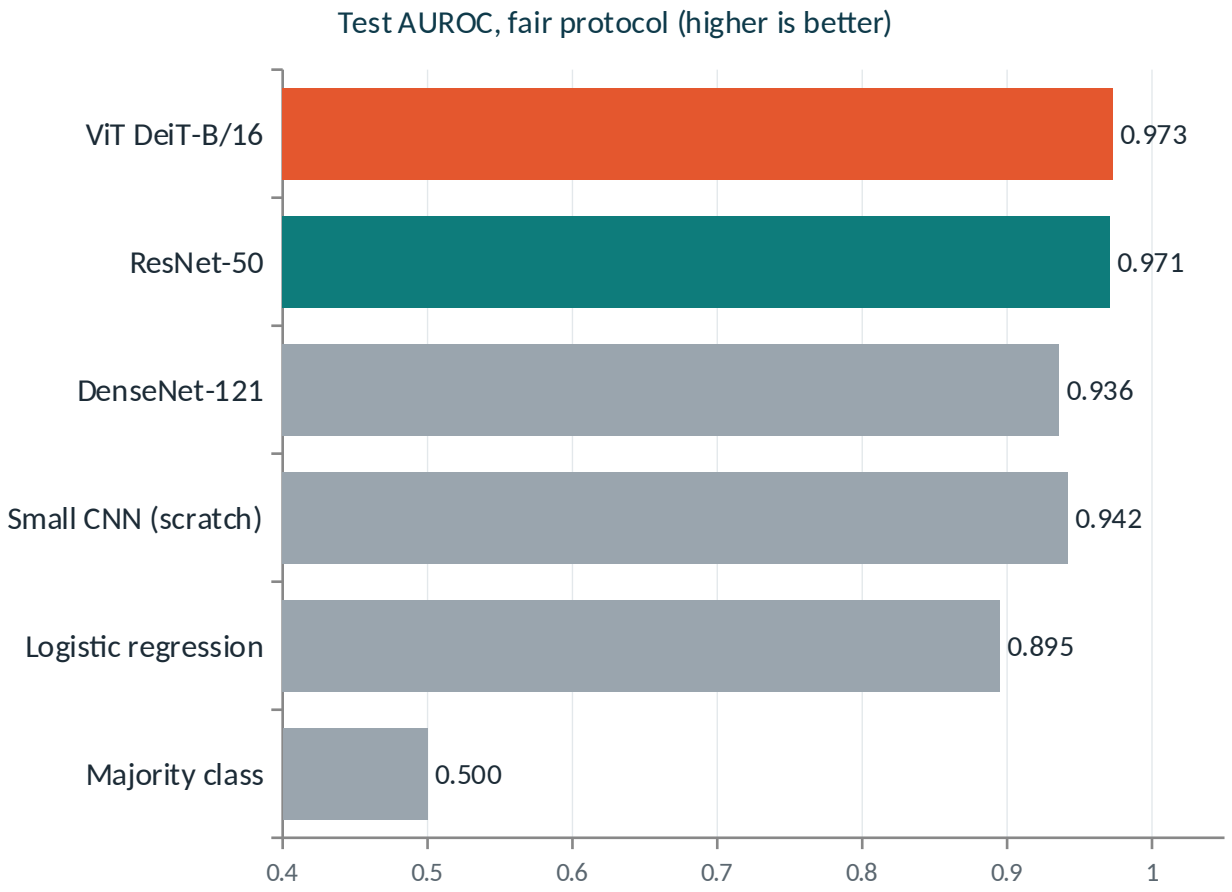
points above the 82.4 % they report for ResNet-50 with a frozen backbone

Read it as

Give the CNN the same treatment as the transformer and the advantage is 0.8 points, inside the seed-to-seed spread. Their comparison measured whether the backbone was trained, and called it architecture.

Result 3: fair protocol, the ViT ties ResNet-50 at 3.7× the size

All six models, patient-grouped split, official test folder, mean of seeds 42 and 43. Our Assignment 1 table with the paper's model added.



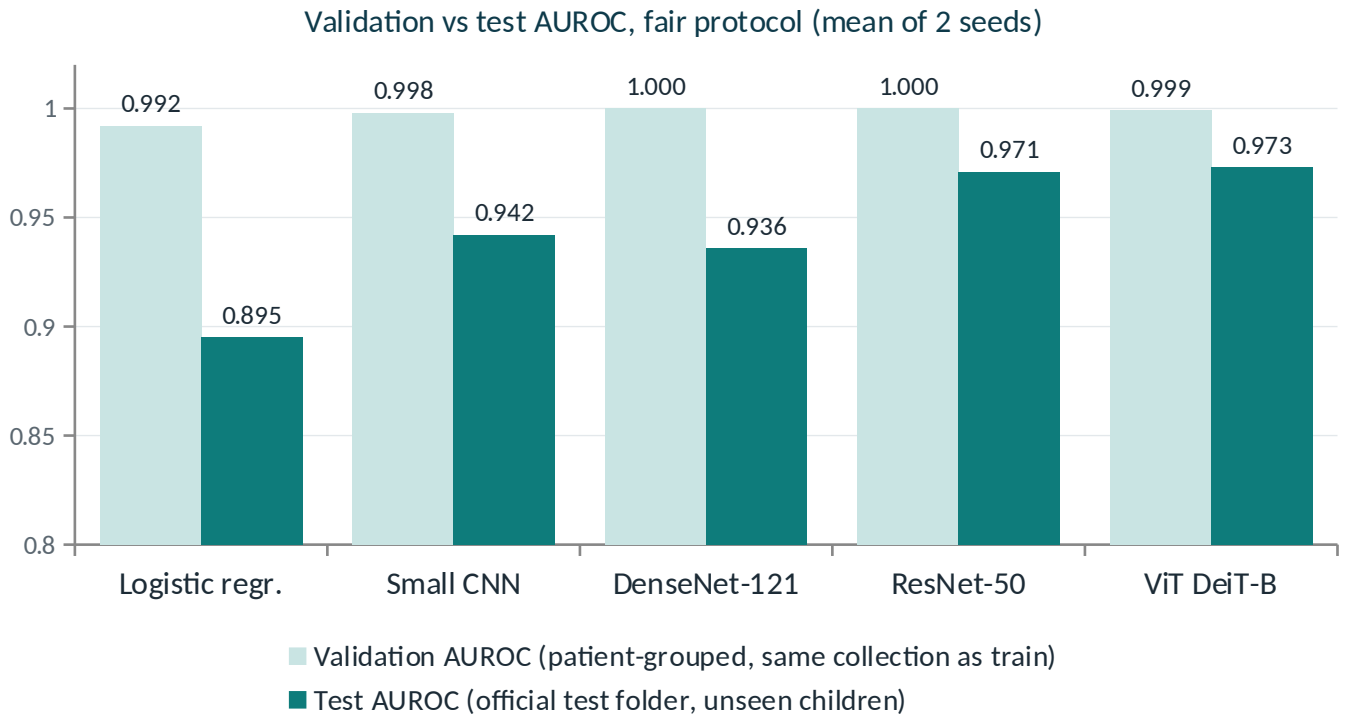
Model	AUROC	Acc. %	Spec.	Params	Train
ViT DeiT-B/16	**0.973 ± .001**	82.4	0.53	85.8 M	9.2 min
ResNet-50	**0.971 ± .001**	86.2	0.64	23.5 M	4.4 min
Small CNN	0.942 ± .006	82.6	0.54	0.39 M	3.4 min
DenseNet-121	0.936 ± .017	82.2	0.53	7.0 M	5.9 min
Logistic regression	0.895 ± .006	77.4	0.43	0.15 M	3.0 min
Majority class	0.500	62.6	0.00	0	0

A tie, at a price

AUROC 0.973 vs 0.971: the seed-to-seed spread is 0.001, so this is a tie. The transformer uses 3.7× the parameters and 2× the training time, and is worse on accuracy and specificity. Sensitivity is ≥ 0.98 for every model; the differences are all in the healthy children.

Why: two properties of the data explain the whole gap

Every model scores ≥ 0.99 AUROC on validation. The drop happens only when the test set is the official folder with unseen children.



Property 1 · Patient leakage

Under their split 58 % of test images have a sibling scan in training. Specificity jumps from 0.53–0.64 to 0.97–0.98 the moment that is allowed. Memorising a child is rewarded as accuracy.

Property 2 · Collection shift

The official test folder is a different collection (A1 audit: different image geometry and class prior). A random 10 % of the training pool never meets it, so their test is easier by construction.

Plus one property of their setup: frozen baselines.

Trained equally, ResNet-50 and ViT are within 0.8 points under their protocol and 0.002 AUROC under ours. Architecture did not move the number; the protocol did.

VERDICT

The claim does not hold up. The protocol, not the architecture, produced the number.

Reproduced

Their 97.6 % → our 98.0 % under their split. Their evidence is real for the question they asked.

Explained

A fine-tuned ResNet-50 gets 97.2 % under the same split. Patient leakage (58 %) and a same-collection test set inflate every model alike.

What survives

"On this dataset the evaluation protocol dominates the architecture." Under a fair protocol the ViT ties ResNet-50 at 0.97 AUROC, at 3.7× the size.

PART 2 · COURSE PROJECT (OPTION 3: AI AGENTS)

Can an LLM write a pneumonia classifier's weights, with no training at all?

Part 1 showed that the evaluation decides what a number means. Part 2 keeps our fair pipeline fixed and changes who sets the weights.

Where it came from

Our critique's next steps were an external test set and a training-size sweep. We took Option 3 instead: in class we met MIT's "Sparks of Weight-Space Programming" (Takehi et al., Nov 2026, blog post), where GPT-6 Astra wrote and edited network weights directly. We have access to Astra.

Research question (one sentence)

Can GPT-6 Astra, writing all 5,569 weights of a small CNN by reasoning alone, beat a random search given the same number of practice evaluations on test AUROC, and how close does it get to the same CNN trained by gradient descent?

What MIT did, and what we copied

Their wins came from tasks with rules the LLM could read. Pneumonia has no source code: only labelled X-rays define it.

MIT mode	What the LLM got	Their result	Our version
Write from scratch	Game rules, source code, network shape; no data	MinAtar policies beat PPO at 100 M steps	**A–D, F: Astra writes all 5,569 weights of a small CNN · G: Astra writes the last layer of an ImageNet ResNet-50**
Edit	A failing trained network + forward passes	Qwen MATH-500: 7.9 → 48.1 %	E: Astra edits the last layer of our trained ResNet-50
Read	Raw weights: "what was this trained on?"	Named the game or language	Not used

Why this is a hard test

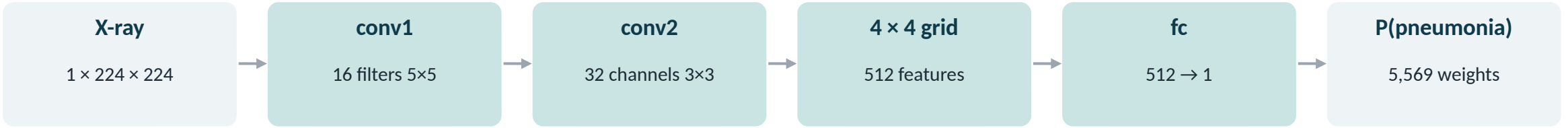
In the games the LLM effectively compiled known rules into weights. Here, whatever it writes must come from what it knows about radiographs and from the feedback we give it.

What makes it fair

Same data, patient-grouped splits, preprocessing and seeds as Assignment 1. Same network for every row. Each rival gets exactly as many practice evaluations as Astra used. The test set is scored once per condition.

Setup: one small CNN, five things Astra could be given

Each condition is a fresh Astra chat. It returns a Python file that builds every weight from constants; no training of any kind.



Condition	What Astra gets	Practice evaluations
A	The task, the preprocessing and the network code. Nothing else.	0 (one reply)
C	+ 16 labelled example X-rays to look at	0 (one reply)
B	+ practice passes: we run its file on 256 practice images and return scores	up to 20
D	+ images and practice passes	up to 20
F	**+ images, practice and a channel report (how every filter responds to each class) · 3 independent chats**	up to 20

Rules (checked by code)

No gradients, no fitting, no search loops: every change is designed and explained.

Only numpy and math, no random numbers, no file access: a static check and an isolated process enforce it.

Dataset not named, no image sizes shown.

Test set scored once.

The practice loop, and the rivals that make it a fair fight

Astra never sees a practice image, only numbers about how its own network did.

Astra writes

a weights file
(by reasoning)

→

We check it

static rules +
isolated process

→

One forward pass

next 256 practice
images, no training

→

We return

AUROC, sens., spec.,
per-image scores
+ channel report (F)

↻ Astra reads the result, decides one change, and repeats; it says FINAL when done (it stopped after 4 passes in every F chat).

A real channel-report line (F, run 2)

conv2 channel 8 (local intensity range): separation 0.061, i.e. clearly higher on NORMAL images.

Astra's next file read only this channel, with weight −1: practice AUROC 0.59 → 0.93.

Rival 1 · matched random search

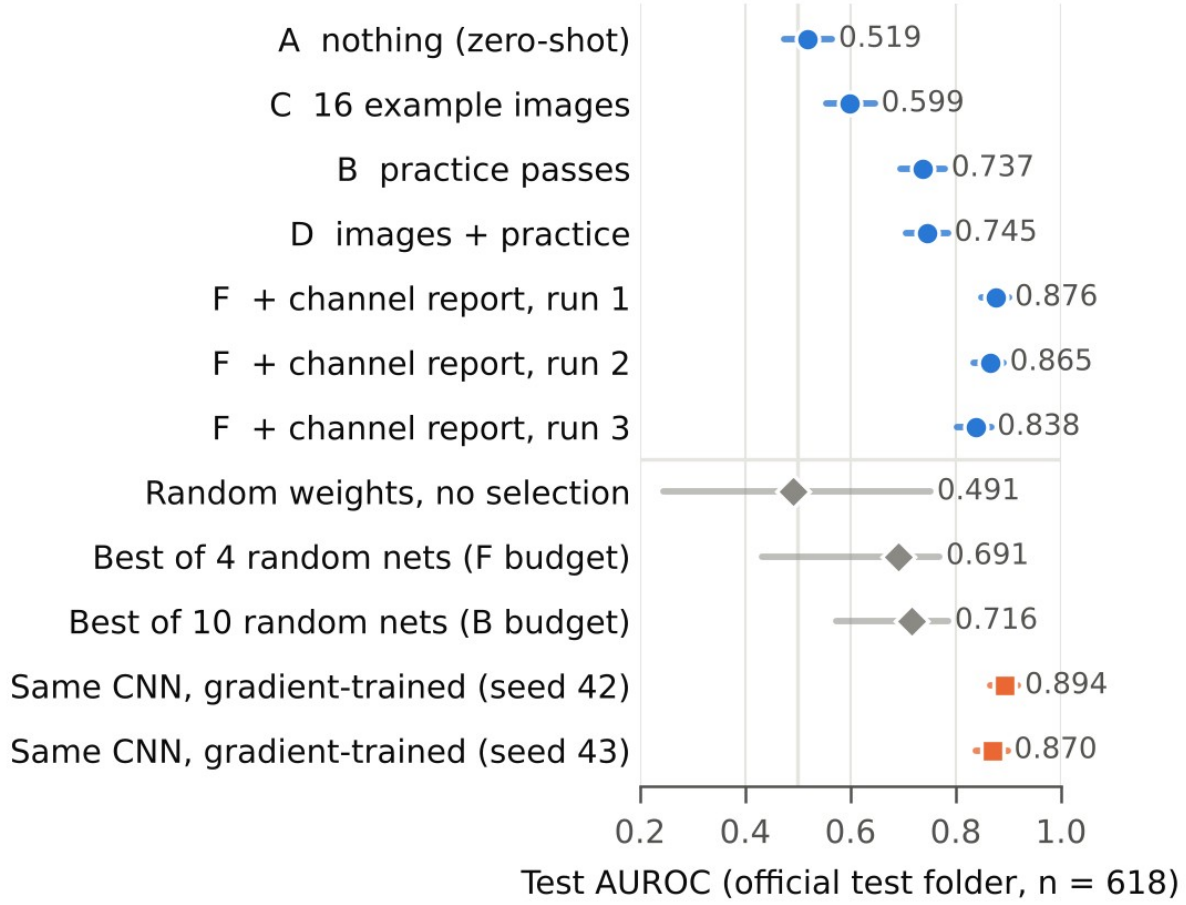
Draw N random networks, score each on the same practice slices Astra saw, keep the best; N = Astra's number of passes. Repeated 100 times.

Rival 2 · gradient training

The same 5,569-weight CNN trained with the Assignment 1 recipe (AdamW, 12 epochs, seeds 42 and 43): 3.3 min on a T4.

Result: the channel report brings Astra close to training

Test AUROC on the 618 official test images (patient-grouped). Bars: 95 % CI, or 5th–95th percentile for random rows.



Bars: 95% bootstrap CI (Astra, trained); 5th-95th percentile over 100-200 draws (random rows).

0.519

nothing given (A): chance

0.737

practice scores (B); images add only +0.008 (D)

0.859

± 0.016, channel report (F), 3 chats

0.882

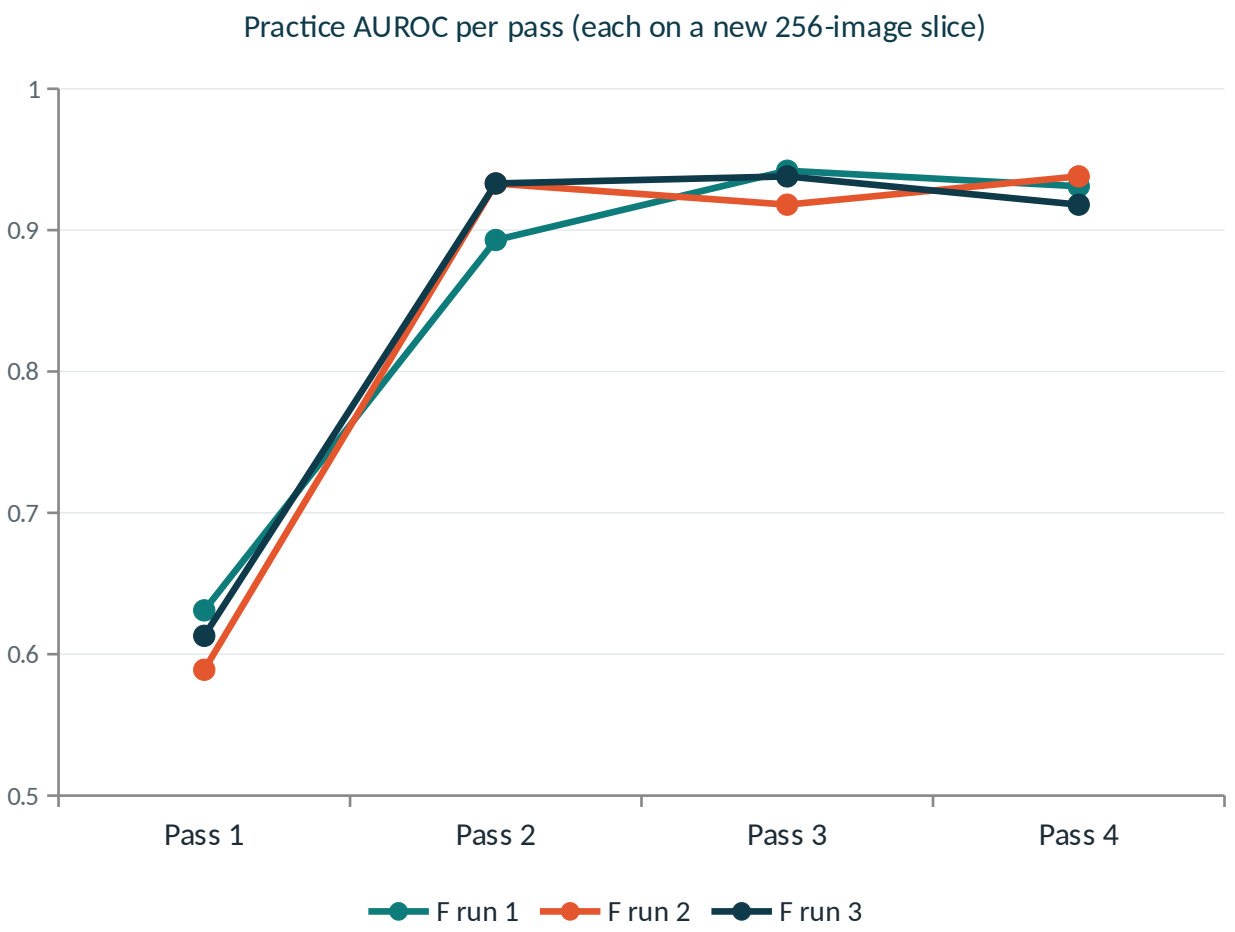
± 0.012, same CNN trained by gradient descent

Read it as

Every F chat beat all 100 matched random searches (best 0.821). With practice alone (B), Astra was not clearly better than random search (28 % of searches matched it).

Three chats, one rule: less fine texture means pneumonia

Each chat started from a different weak design; the first channel report turned each one around.



What changed after pass 1

Astra saw that its own texture channels (edge strength, local contrast) were consistently higher on NORMAL images, and rebuilt its output around them with round weights (−1, −0.5).

How simple the final rule is

Run 3's final network reads one channel, average edge strength, with weight −1 per grid cell and bias +1.25. Healthy lungs show more fine structure.

Honest framing

No weight was fitted, but the report is label-informed: it tells Astra which of its own hand-built filters separate the classes. That is feature selection by reasoning: no training, but not data-free.

Stronger features (G): Astra writes ResNet-50's last layer

Backbone trained only on everyday photos, never on X-rays. Astra writes 2,048 weights + 1 bias; same practice loop, one report row per feature.

Last layer obtained by	Test AUROC [95 % CI]	Features used	Size-matched AUROC
Astra · 17 practice passes + feature report	**0.902 [0.877, 0.922]**	26	0.724
Selection rule: top features by separation (no LLM, no gradients)	0.883 [0.853, 0.905]	32	0.669
Best of 17 random layers (100 repeats, median)	0.678 (max 0.800)	2,048	0.571
Linear probe, trained (logistic regression)	**0.947 [0.926, 0.963]**	2,047	0.862

Better features help

0.859 on the small CNN → 0.902 here. Beats random search and, narrowly, the mechanical rule (CIs overlap).

Training wins more clearly

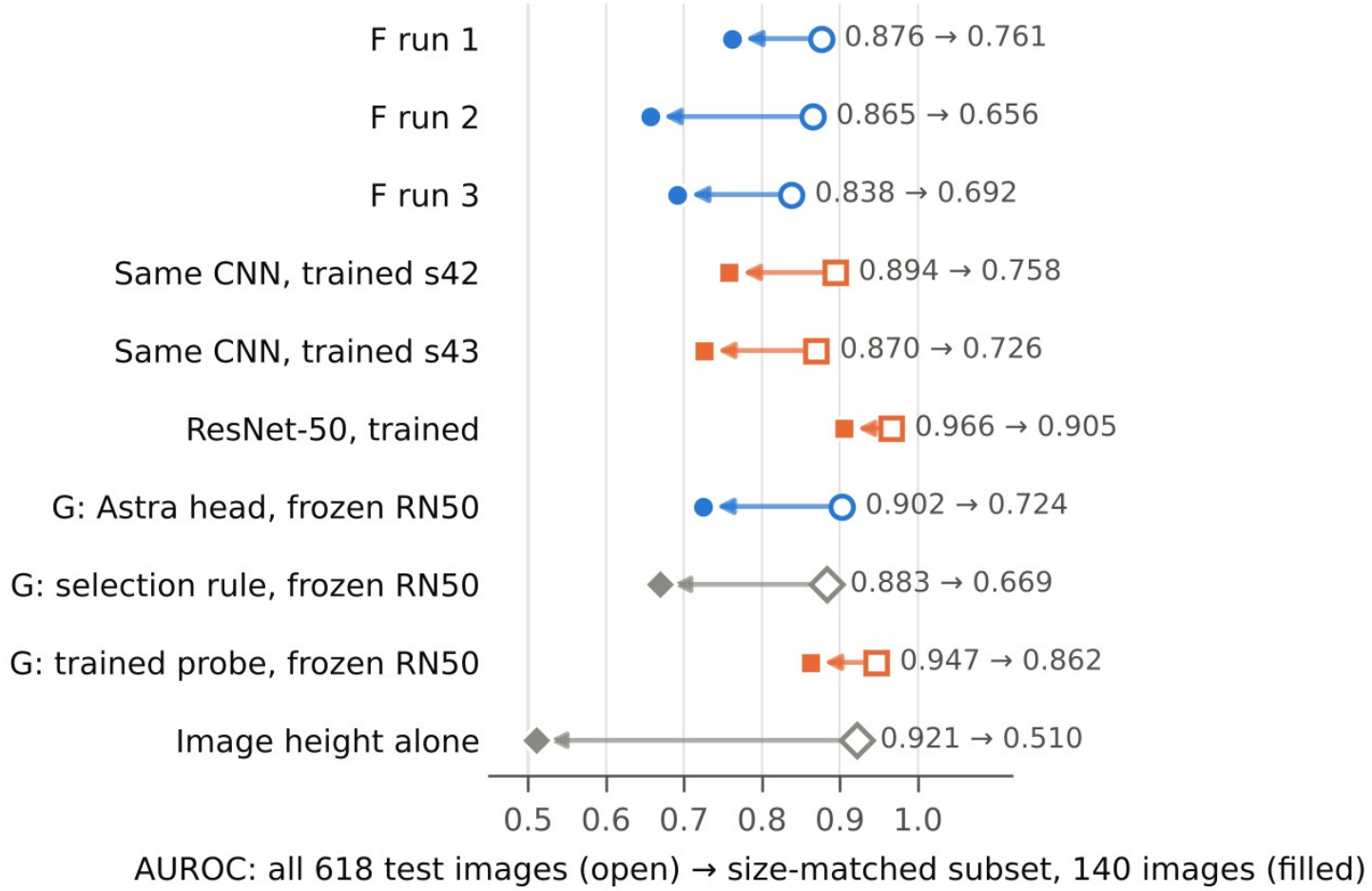
Gap to the trained probe: 0.045, twice the gap on the small CNN. Hand-picking 26 of 2,048 unnamed features leaves signal unused.

Astra knew

It predicted ≈ 0.96 and warned that 17 looks at practice data make that optimistic. Practice ran 0.96–0.98; test 0.902.

Robustness: part of every score is image size

Astra's rules lean on it more than trained weights do. A1 found that original image height alone predicts the label (AUROC 0.92); we re-scored on 140 size-matched test images.



Everyone drops

On the matched images, height alone falls to 0.51: the check works. Every model loses AUROC, so the dataset itself rewards this shortcut.

Astra drops more

Small CNN: Astra 0.859 → 0.703 vs trained 0.882 → 0.742.

ResNet-50 features: Astra 0.902 → 0.724 vs trained probe 0.947 → 0.862.

Why (a property of the data)

Pneumonia files are smaller, so after resizing they show less fine texture. A 'less texture' rule reads both disease and file size.

Edit mode: with the feature report, Astra improved our ResNet-50

Astra changed the last layer (2,048 weights + bias) of our fine-tuned ResNet-50. Practice: one half of the test folder (306 images); scored on the other half (312 different children).

Last layer of our fine-tuned ResNet-50	Test AUROC [95 % CI]	Balanced acc.	Size-matched
Unedited	0.954 [0.925, 0.977]	0.745	0.879
Threshold moved on the practice images	0.954	0.914	0.879
E · Astra edit, scores only	0.954	0.814	–
H · Astra edit + per-feature report	**0.976 [0.959, 0.990]**	0.903	0.922
Re-trained on the same 306 images (logistic regression)	**0.985 [0.974, 0.995]**	0.941	0.925

E: scores only

Astra could only move the cut-off: AUROC stayed 0.954, and a plain threshold move did better.

H: with the report

It replaced the layer with ±1 votes on 49 features that separate this collection: AUROC 0.954 → 0.976. Retraining still wins (0.985).

Not a size shortcut here

Size-matched 0.922 vs 0.925 for retraining: fine-tuned features already track the disease (72 images, indicative).

CONCLUSION · BOTH PARTS

In both parts, the baseline decided what the number meant

Part 1 · Paper critique

The ViT's 97.6 % reproduces under a random image split, and a fine-tuned ResNet-50 gets 97.2 % under the same split. With patient grouping and the official test folder, both tie at 0.97 AUROC. The protocol, not the architecture, produced the claim.

Part 2 · Project

Yes, with the right feedback: GPT-6 Astra wrote a classifier with no training that beat every matched random search (0.859; 0.902 on ImageNet features), and lifted our trained ResNet-50 from 0.954 to 0.976. Gradient training with the same information still wins every time (0.882; 0.947; 0.985).

Limitations, and what we would do next

Everything here is one dataset and one LLM; the loop was run by hand.

Limitations

F: 3 chats; other conditions: 1. Manual loop: one log reset, one accidental re-run (documented; no extra information to Astra). In G, Astra could read the reports with a tool and saw only their top and bottom. Size-matched set: 140 images.

Project, next

Automate the loop; add a second LLM; normalise image size before Astra sees any feedback, to remove the shortcut; report on groups of features in G, so the LLM can use more of them; test on another pediatric hospital.

Critique, next

Close the budget and preprocessing gaps (30 epochs, plain resize, a third seed) and test the ViT where it might win: an external pediatric set and a training-size sweep.

Our question for another group

"In the paper you critiqued, were the baselines trained with the same budget and the same number of trainable parameters as the proposed model, and how would you check that from the paper alone?"

Backup: every number in one table

Mean ± std over seeds 42 and 43. n/r = not reported by the paper. Result files: results/*.json in the repository.

Model	Protocol	AUROC	Accuracy %	Sens.	Spec.	Params	Train
ViT DeiT-B/16 (reported)	Singh et al., random by image, n=586, 1 run	0.966	97.6	0.949	0.981	85.8 M	n/r
ViT DeiT-B/16 (ours)	Singh et al. protocol in our pipeline, n=583	0.997 ± .002	98.0 ± 0.6	0.984	0.972	85.8 M	9.0 min
ResNet-50 (ours)	Singh et al. protocol in our pipeline, n=583	0.995 ± .004	97.2 ± 0.4	0.969	0.978	23.5 M	5.0 min
ResNet-50 (reported)	Singh et al., backbone frozen	n/r	82.4	n/r	n/r	0.2 M trainable	n/r
DenseNet-121 (reported)	Singh et al., backbone frozen	n/r	91.8	n/r	n/r	0.1 M trainable	n/r
ViT DeiT-B/16 (ours)	A1: patient-grouped, official test, n=618	0.973 ± .001	82.4 ± 2.7	0.999	0.530	85.8 M	9.2 min
ResNet-50 (A1)	A1 protocol	0.971 ± .001	86.2 ± 3.4	0.997	0.636	23.5 M	4.4 min
DenseNet-121 (A1)	A1 protocol	0.936 ± .017	82.2 ± 2.7	0.999	0.526	7.0 M	5.9 min
Small CNN (A1)	A1 protocol	0.942 ± .006	82.6 ± 0.3	0.996	0.541	0.39 M	3.4 min
Logistic regression (A1)	A1 protocol	0.895 ± .006	77.4 ± 4.2	0.982	0.426	0.15 M	3.0 min
Majority class (A1)	A1 protocol	0.500	62.6	1.000	0.000	0	0

Backup · weak point 3: numbers that do not reconcile

Small inconsistencies are not fatal on their own. Together they show the evidence was not checked carefully.

Table 3 says

Test set: 330 pneumonia, 256 normal (586 images).

≠

The confusion matrix says

TP 152 + FN 8 = 160 pneumonia; TN 420 + FP 6 = 426 normal.

Same test set, different composition.

Also

F-score 0.952 (Table 5) vs 0.950 (Table 6) · cross-entropy vs binary cross-entropy · CI 96.2–98.9 % stated, 96.4–98.9 % by their own Eq. 7 · 5,863 images in text, 5,856 in tables

One run

No seed, no cross-validation, no variance. Test accuracy is tracked every epoch next to validation ("improves along with test accuracy"), which risks selecting on the test set.

Accuracy as headline

On a set that is 56–73 % pneumonia, a model that always says 'pneumonia' already scores the majority share. AUROC or balanced accuracy would be the honest lead metric.

Limitations and reproducibility

Limitations: one sentence ("data scarcity and the need for real-world validation"). No code, no split files, no seeds. The experiment can only be approximated.

Backup · project: every number in one table

Official test folder, n = 618. Astra time = first to last practice pass, including manual copy-paste. Files: results/written_table.md, head_resnet50.md, robust_size.md.

Weights obtained by	Test AUROC	Sens.	Spec.	Size-matched	Params	Time
A · Astra, zero-shot	0.519 [0.47, 0.57]	0.74	0.23	–	5,569	1 reply
C · Astra, 16 images	0.599 [0.55, 0.65]	0.89	0.14	–	5,569	1 reply
B · Astra, 10 practice passes	0.737 [0.69, 0.78]	0.59	0.82	0.677	5,569	72 min
D · Astra, images + 11 passes	0.745 [0.71, 0.79]	0.69	0.72	0.693	5,569	42 min
F · Astra + channel report, runs 1 / 2 / 3	0.876 / 0.865 / 0.838	0.91	0.50	0.761 / 0.656 / 0.692	5,569	20–30 min
Best of 4 / 10 random nets (median)	0.691 / 0.716	–	–	–	5,569	seconds
Same CNN, trained (seeds 42, 43)	0.894 / 0.870	0.97	0.49	0.758 / 0.726	5,569	3.3 min
G · Astra head, frozen ImageNet ResNet-50	0.902 [0.88, 0.92]	0.94	0.59	0.724	2,049	17 passes
G · trained linear probe, same features	0.947 [0.93, 0.96]	0.99	0.53	0.862	2,049	< 1 s
ResNet-50 fine-tuned (A1)	0.971 ± 0.001	1.00	0.64	0.905	23.5 M	4.4 min
H · Astra edit of fine-tuned ResNet-50 (n = 312)	0.976 [0.96, 0.99]	0.98	0.83	0.922 (72 img)	2,049	7 passes
H · layer re-trained on the 306 practice images	0.985 [0.97, 1.00]	0.97	0.91	0.925 (72 img)	2,049	< 1 s

Backup · how we kept it honest

Every deviation is logged in astra/README.md; prompts, transcripts, weight files and practice logs are in the repository.

Could a new chat see earlier runs?

Model weights do not change between chats. Transcripts log every file Astra opened: each run read only its own prompt and outputs. Each run started from a weak first design (0.59–0.63).

Manual-loop slips

F run 1: Colab reset; passes 3–4 reused the slices of passes 1–2. F run 3: one accidental re-run never shown to Astra. Rivals got the larger budget.

Tool use

A–F: no code run. E: Astra computed weight summaries (mean, RMS) from the weight file only. G: a read-only tool for the long reports (36 Get-Content reads, no computation).

Design changes made before Astra saw anything

E: practice moved from the validation set (model already 0.9995 there) to one half of the test folder; scored on the other half. Practice output never includes image size (A1 showed it leaks the label).

What the channel report really is

Label-informed feature selection: it tells Astra which of its own filters separate the classes. No weight is a formula of the statistics (prompt rule + transcript check), but the method is not data-free, and we say so.